

Cognitive and Neural Science

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The Neuroscience Approach

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Outline

- The neuro science perspective
- The human brain
- Methodology in neuroscience

Approaches in Cognitive Science: The Neuroscience Approach

- The general study of the brain and endocrine system is called neuroscience.
- The attempt to explain cognitive processes in terms of underlying brain mechanisms is known as cognitive neuroscience.
- Neuroscience, first and foremost, provides a description of mental events at the implementational level.
- It attempts to describe the biological “hardware” upon which mental “software” supposedly runs.

Neuroscience is the intricate study of the nervous system, striving to uncover the principles that govern brain function, cognition, behavior, and disease.

The Neuroscience Perspective

- Cognition is the process and activity by which the brain and nervous system, producing the mind.
- Any animal with a brain and nervous system has a certain degree of mind.
- Neuroscience tries to understand and explain how the human brain works.
- It includes all disciplines related to anatomy and nervous system in humans and other animals as well as its function.

The Neuroscience Perspective

- There has been a fairly recent trend in neuroscience toward the integration of biology with cognition, called **cognitive neuroscience or neuropsychology**.
- The goal of this discipline is to **explicate the structures and physiological processes** that underlie specific cognitive functions and operations:
 - **Visual object recognition, attention, memory, and problem solving**

The Big Picture: Brain Anatomy

- The ancient Greeks were among the first people to study the brain
- Tried to understand the role of the brain and how it worked and to explain neural disorders.
- The human brain is far more plastic than what we had imagined.
 - Some of its functions seem to be transferable to areas of the brain not used for those functions under ordinary circumstances

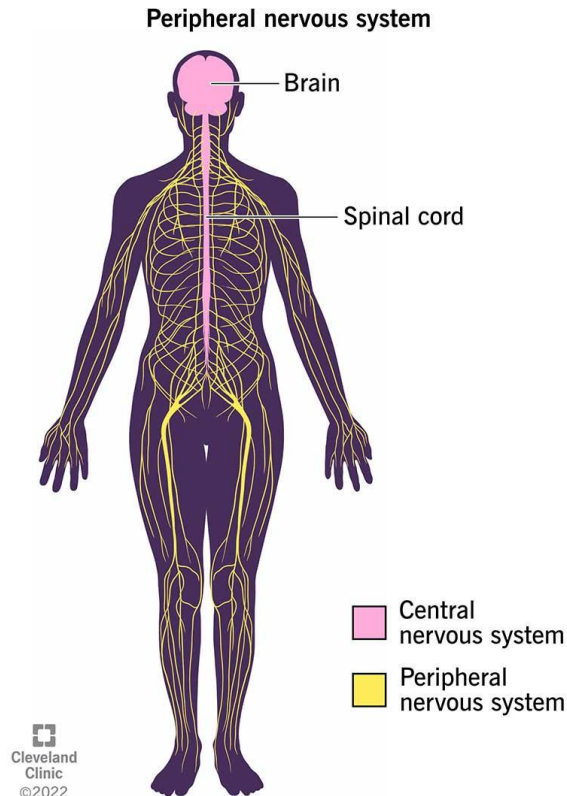
“One of the most intriguing aspects our study of human brain is that we have learned and continue to learn about human brain by using it to study itself”

The Nervous System

- The nervous system is a **marvel of complexity**, comprising billions of neurons and glial cells interconnected in dynamic and adaptive networks.
- Consists of the brain, spinal cord and peripheral nerves

Has two parts:

- **Peripheral nervous system:** made up of afferent nerve fibers— carry signals from the senses to the brain and efferent nerve fibers ... carry messages from the brain to muscles
- **Central nerve system:** consists spinal cord and brain



Src: [Cleveland Clinic](https://my.clevelandclinic.org/health/body/14027/nervous-system)

Parts of the Brain

Parietal lobes consists of several areas which could be used for **recognition and recall of form, texture, and weight**; awareness of moving objects

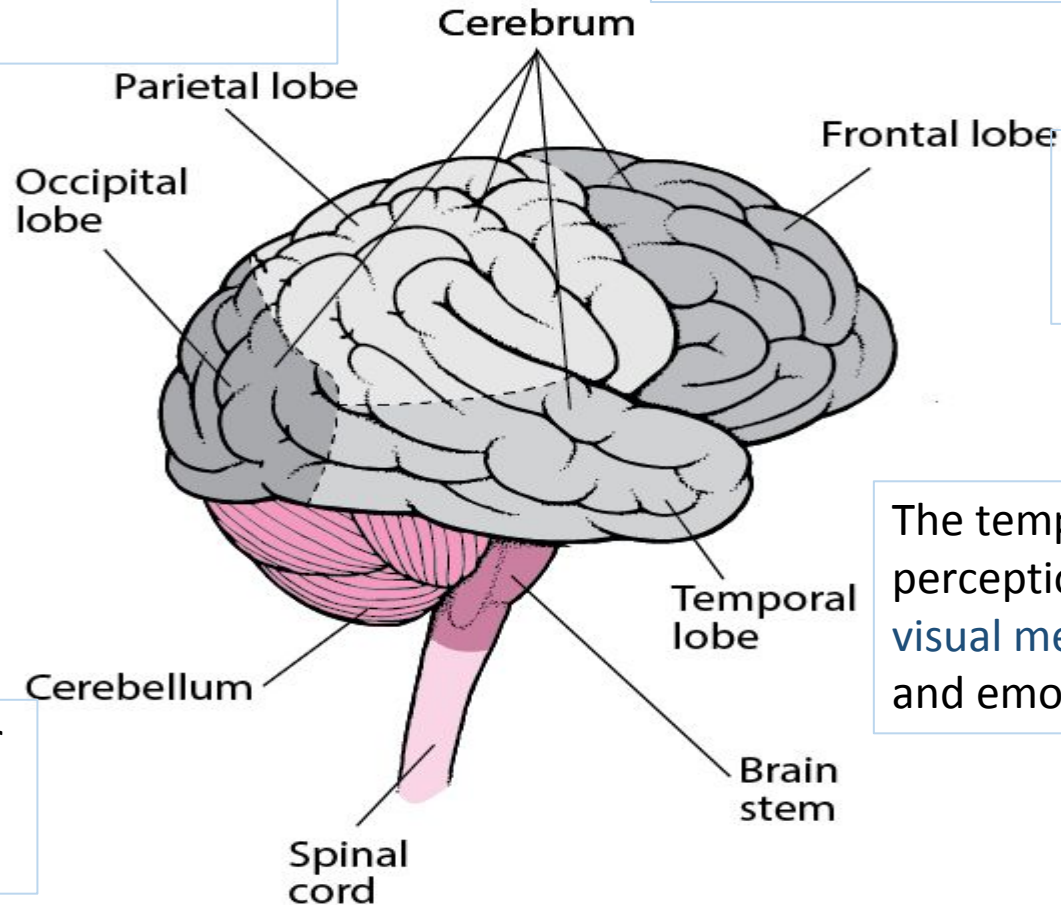
The cerebrum contains the major lobes of the brain and is responsible for **receiving and giving meaning to information from the sense organs**.

The occipital lobes contain the primary visual cortex and visual association areas

Frontal lobes are essential for **planning and executing learned and purposeful behaviors**

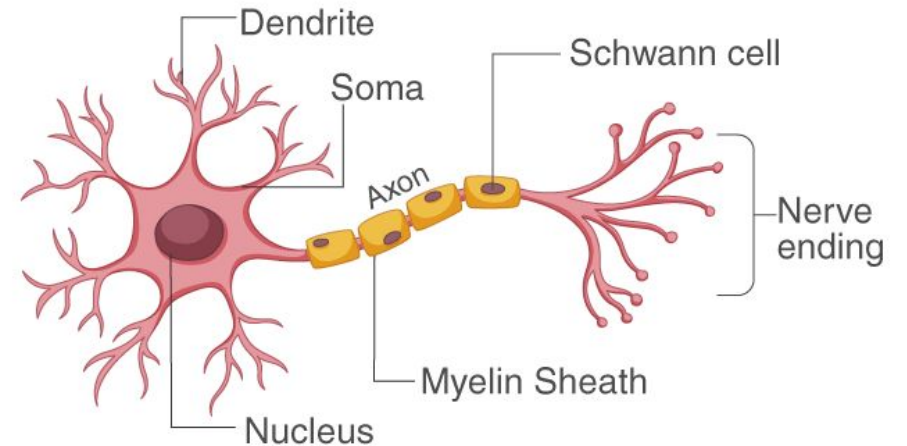
Cerebellum is responsible for coordination and balance

The temporal lobes are integral to auditory perception, receptive components of language, **visual memory, declarative (factual) memory, and emotion**



Parts of the Brian...

- The cells that make up the nerve system are called **neurons**.... a human brain made up of a billion of neurons
- Neurons vary in size, **from very small to 3 feet long**, and made of three parts
 - **Soma**:- cell body, contains the cell nucleus and genetic materials (DNA & RNA)
 - A membrane surrounding the soma receives information from other cells
 - **Dendrites**: branching systems and think ribbons that extends to the body
 - **Axons**: a single and longer branch that originates from the soma



Parts of the Brian...

- Three types of neurons:
 - **Sensory neurons** are coupled to receptors specialised to detect and respond to different attributes of the internal and external environment.
 - **Motor neurons** control the activity of muscles, are responsible for all forms of behaviour including speech.
 - **Interneurons** mediate simple reflexes as well as being responsible for the highest functions of the brain

Parts of the Brian...

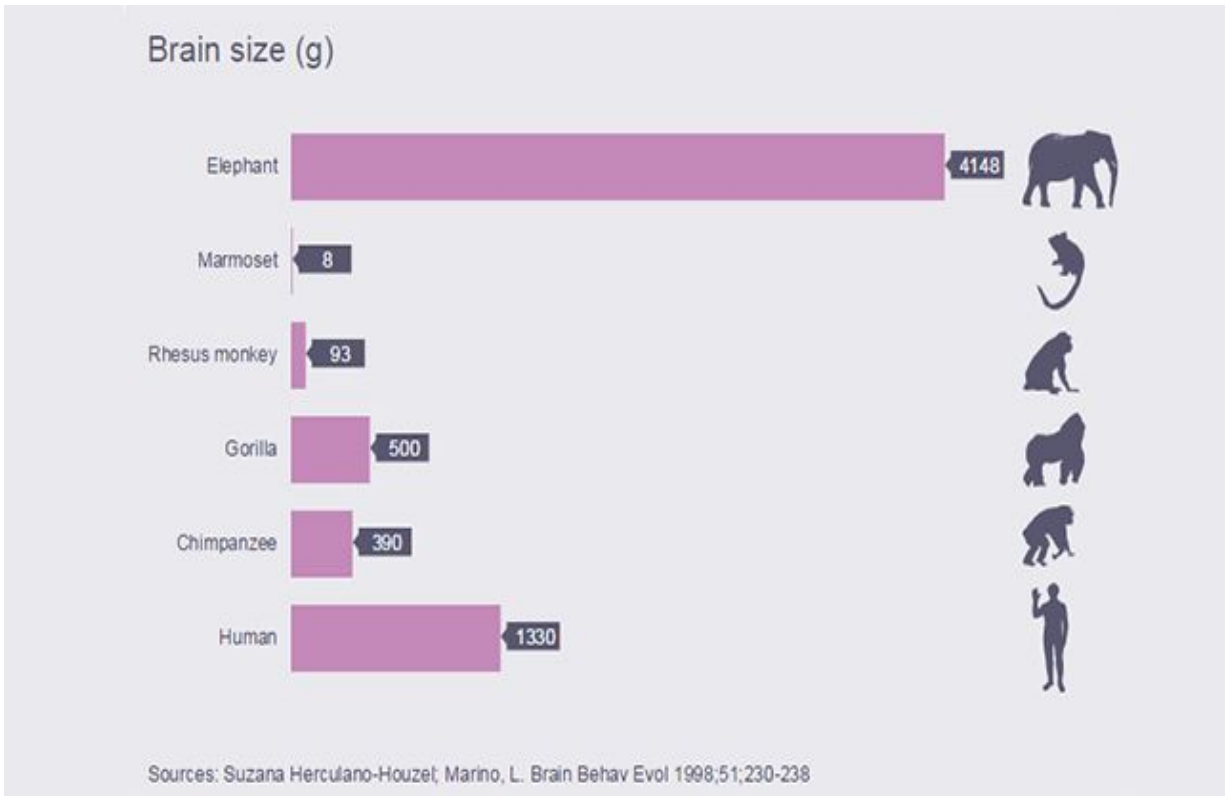
- Nerve impulses produced from neurons
 - Are electronic charges that travel along the axon to the axon terminal
- The axon terminals are the transmitter of information to the membrane of other neurons across a small space called synaptic cleft.
- Despite the fact that much is still unknown about how the human brain functions, there is a good deal of understanding about it.

How large is the human brain?

How does the human brain compare to those of other mammals?

Human Brain

- The human brain has tripled in size from the last 6 to 8 million years
- Although most brain areas have increased in size over time.... The **neocortex** has increase most in human and other species.
 - Involved in higher cognitive processes, e.g., thinking and decision making



Does brain size correlate with intelligence?

Human Brian

- Bullmore and Sporns (2012) **two major principles** influencing human brain organization:
 - The **principle of efficiency**: the ability to integrate information across the brain High efficiency achieved through numerous connections in the neurons.
 - The **principle of cost control**: cost minimized with short distance connections
- Why is the brain energy consumption so high when we focus on complex problems?

Human Brian

- The increase in brain activity during task performance compared to resting state is typically 5%.
- Our brain use so much energy since it is nearly active all the time

Human Brain

- Each neuron can make about 1000 connections— used **100 trillion data points** (100 terabytes of information)

- Many agree on **10 terabytes** of brain storage capacity which is equivalent to **470 million books**

- **Working memory vs long term memory**

Understanding the memory functions is crucial to understand all other cognitive functions.

- There are some functional specialization within the human brain
 - E.g areas in the occipital cortex mostly devoted to basic visual processing
- **However, flexibility is one of the most impressive feature of human brain**

Human Brian

“Brain regions that respond selectively to specific kinds of information content are described as functionally specialized, whereas those that respond to a broad range of task demands are considered functionally diverse or flexible.”

Human Brian

- The human brain display greater neuroplasticity even when 50% of the brain is destroyed.
 - We can contrast the neuroplasticity with AI systems?
- Language capacity
 - Communication is an obvious human cognitive advantage

The ability to reorganize and reshape itself in response to learning, environmental changes, or damage occurs at multiple levels

BASIC BRAIN PROCESSES

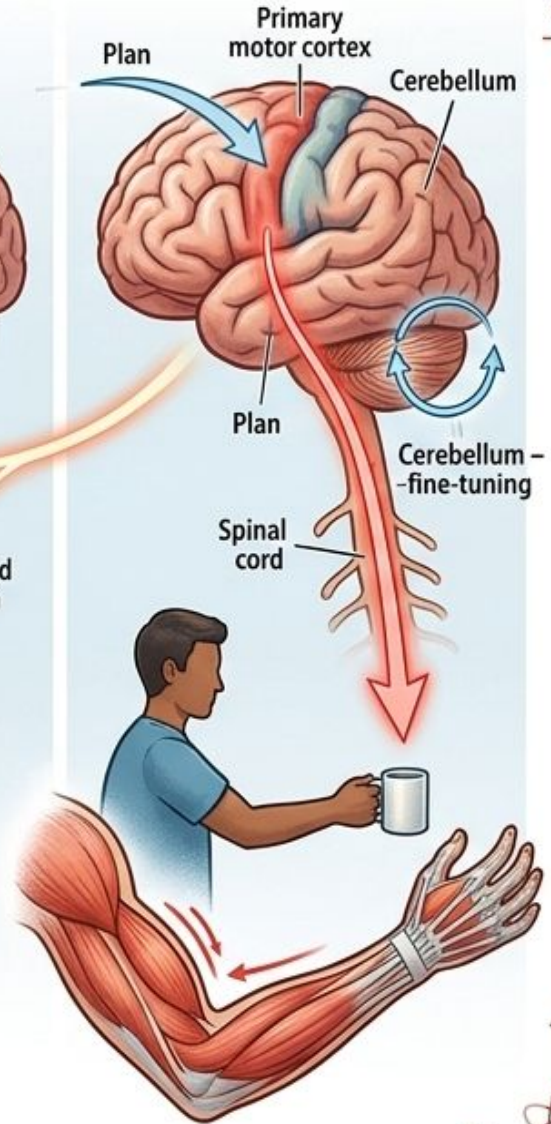
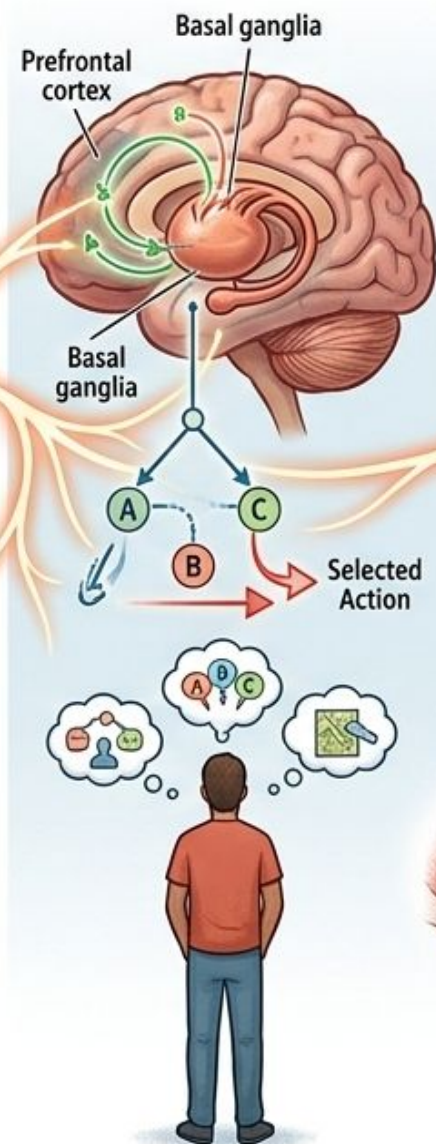
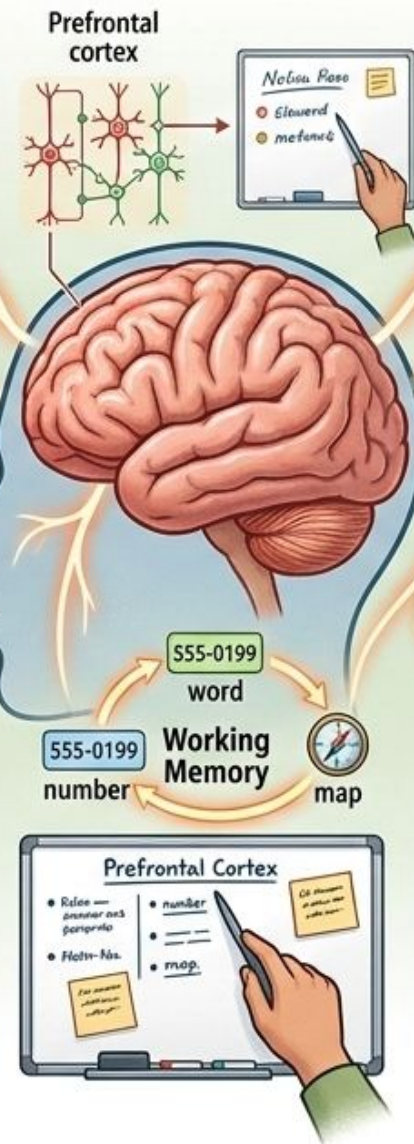
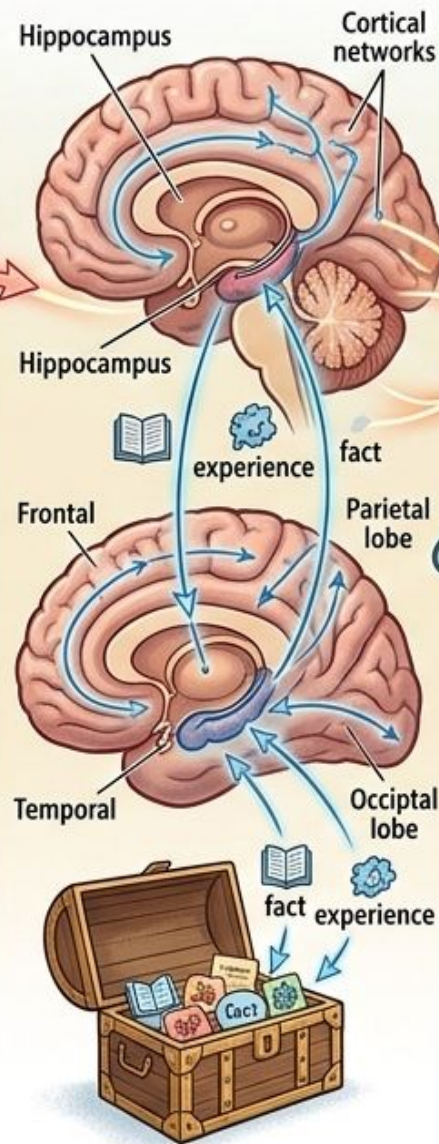
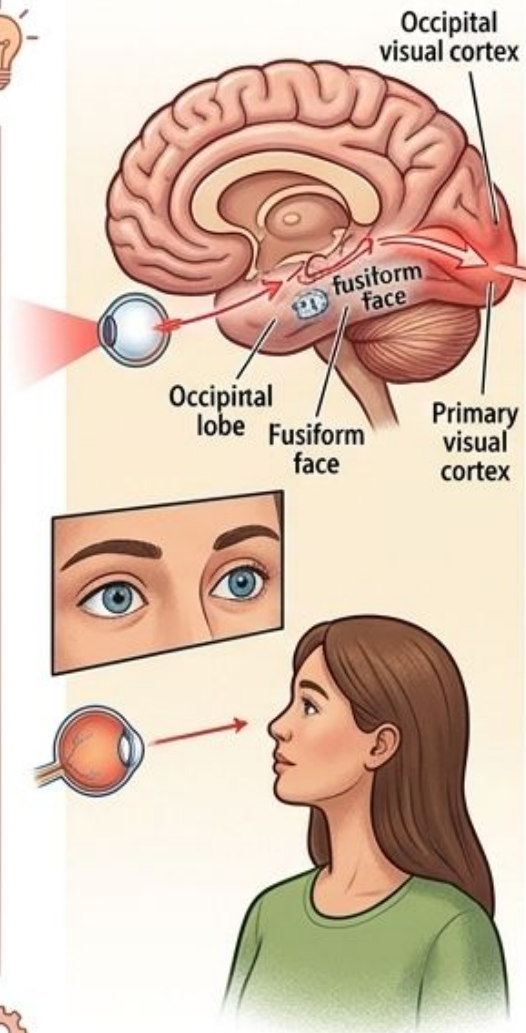
Visual Recognition

Long-term Memory

Short-term Memory

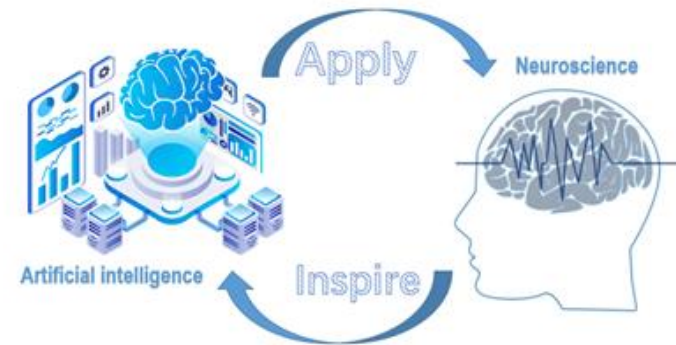
Action Selection

Motor Control



Neuroscience and AI

- These two areas of study influence and stimulate each other.
- The amalgamation of AI and neuroscience shows great prospects in the development of treatments for neurological disorder.
- The use of AI we can learn more about the human brain and its cognitive abilities and it will help us in finding answers to the age-old questions regarding cognition, memory and immortality.
- AI serves as a conceptual bridge, linking neuroscience's understanding of biological intelligence to computational frameworks.



Source: [Frontiers](#)

“This bidirectional exchange has fueled breakthroughs such as dynamic connectivity mapping, real-time neural decoding, and closed-loop brain–computer systems that adaptively respond to neural states.”(Onciul et al., 2025)

Neural Networks & Artificial Brains

- The real brain consists neurons, blood vessels and fluid- filled ventricles are made of lipid membranes, proteins and a great deal of water.
- Mathematicians, physicists, engineers and computer scientists think about the brain by writing equations, making computer models and even hardware devices that mimic the real neurons inside our heads.

Neural Networks & Artificial Brains

- “Our brain contains roughly 100 billion nerve cells, 3.2 million kilometers of ‘wires’, a million-billion connections, all packed into a volume of 1.5 liters, but weighing only 1.5 kg and consuming a mere 10 watts.”
- If we tried to build such a brain using silicon chips, it would consume about 10 megawatts, i.e. enough electricity to power a town.
- The challenge is to discover how brains operate so efficiently and economically, and to use similar principles to build brain-like machines.

Comparing AI and humans with respect to general skills and intelligence

What's the next step for modelling the brain?

- Computational models have been developed for many different systems in the brain over the past ten years.
- The holy grail now is putting them together... developing a multi-system model.
- “It is a huge challenge ... we do not really know how these systems work together in the human brain, and finding out involves a lot of collaboration with neuroscientists in what are currently considered separate fields”

Methodology in Neuroscience

- In neuroscience, the methodologies used to understand how the brain functions are generally classified into three strategic approaches:
 1. **The investigation of brain damage:** This category focuses on lesion studies, where researchers observe changes in behavior or cognitive ability after a specific part of the brain is damaged.
 2. **The recording of brain activity in healthy subjects:** Instead of looking at what happens when the brain is broken, this category monitors the brain while it is working in healthy subjects.
 3. **Direct stimulation of the brain itself:** more proactive; it involves turning on or modulating specific brain areas to see the immediate effect on the subject.

Techniques for the Study of Brain Damage

- Researchers examine brain damage and its effects in people that have come about as a result of accident.. Case study
- Localized brain damage, typically resulting from accidents, strokes, or tumors, to understand the specific functions of that region.
- By documenting the behavioral deficits that follow, scientists can infer the role the damaged area played in healthy cognition.
- Areas of the brain are interdependent.
- The effects of damage to one area could have a variety of functional interpretations.

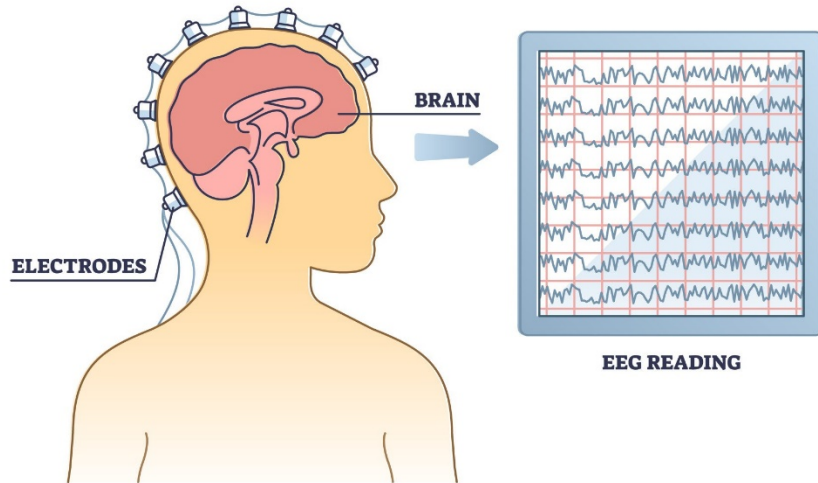
AI as a Tool for Neuroscience:- Brain Recording Techniques

- Neuroimaging has revolutionized how researchers study the brain.
- **Single-cell recording:** a very fine microelectrode is inserted into either a single neuron or the extracellular fluid adjacent to it.
 - Changes in that cell's electrical conductivity or its rate of firing can then be measured
- **Multipleunit recording:** a larger electrode is used to measure the collective electrical activity of a group of neurons.
- If one wants to investigate what the brain as a whole is up to, he or she can choose from a variety of other tools

AI as a Tool for Neuroscience:- Brain Recording Techniques

- **Electroencephalogram (EEG)** is a recording of the brain's gross electrical action

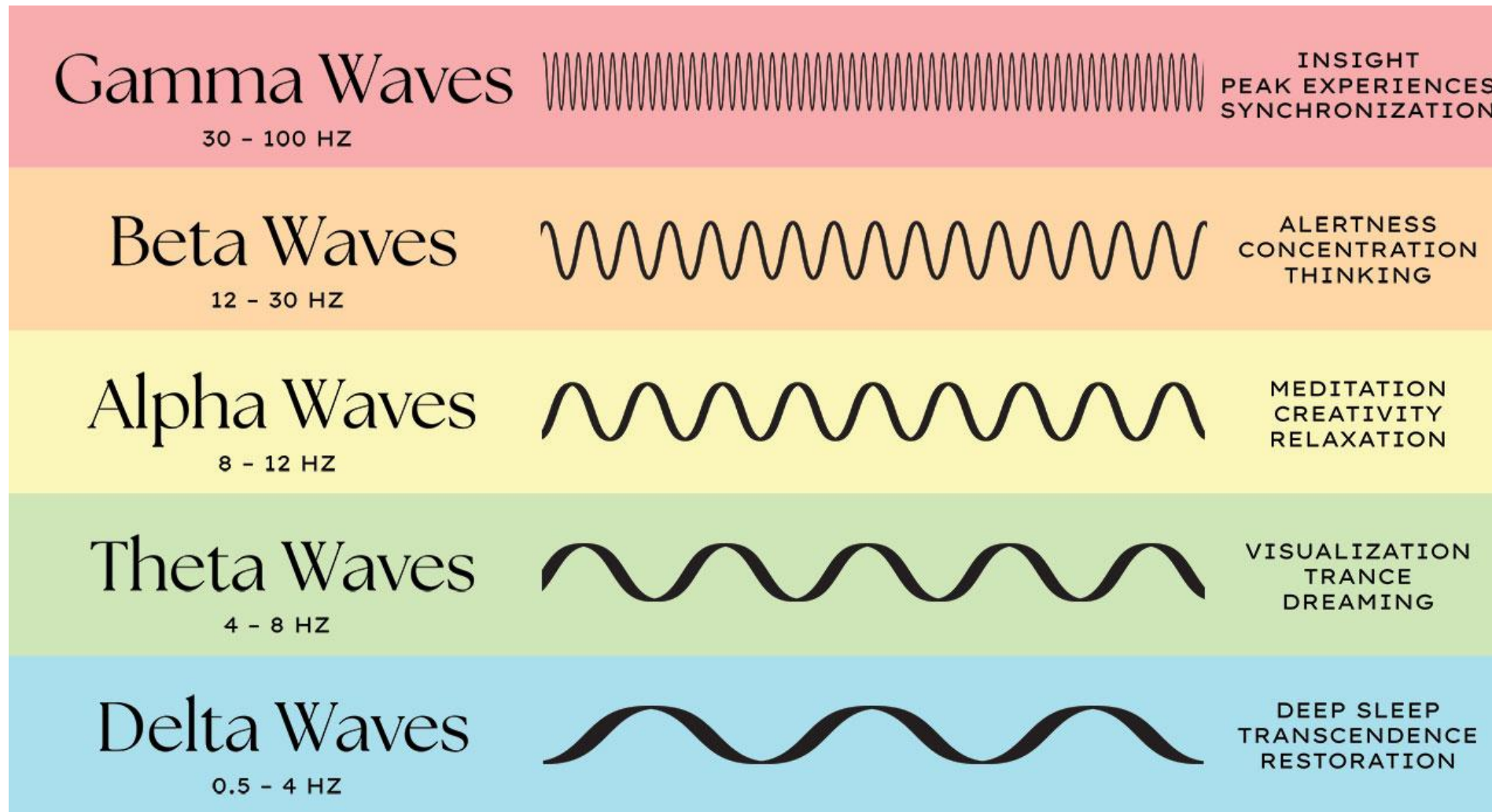
ELECTROENCEPHALOGRAPHY



Key Characteristics of EEG

- It does not record the firing of a single neuron; rather, it captures the summed post-synaptic potentials from thousands or millions of neurons acting in synchrony beneath the electrode.
 - EEG is highly valued for its exceptional temporal resolution, meaning it can record changes in brain activity down to the millisecond.
 - The recording displays rhythmic waves (such as Alpha, Beta, Theta, and Delta) that change based on a person's state of consciousness, such as being awake, relaxed, or in deep sleep.
- The EEG can also be used to measure a subject's brain activity that is in response to his or her experience of a particular event.

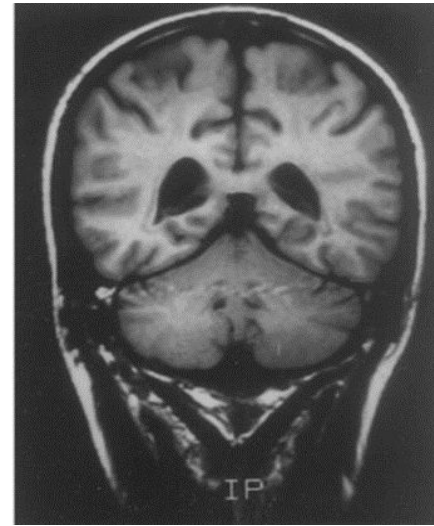
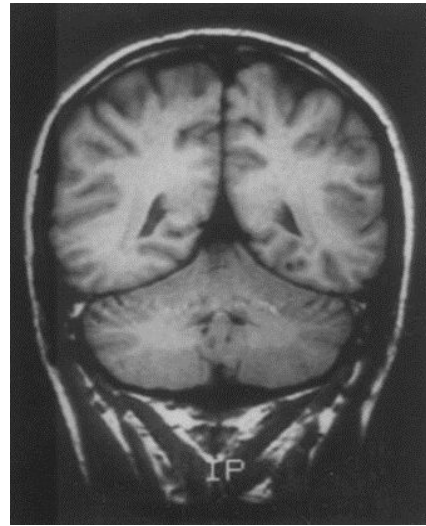
AI as a Tool for Neuroscience:- Brain Recording Techniques....



Source: [Understanding Brain Waves: Beta, Alpha, Theta, Delta + Gamma](#)

AI as a Tool for Neuroscience:- Brain Recording Techniques

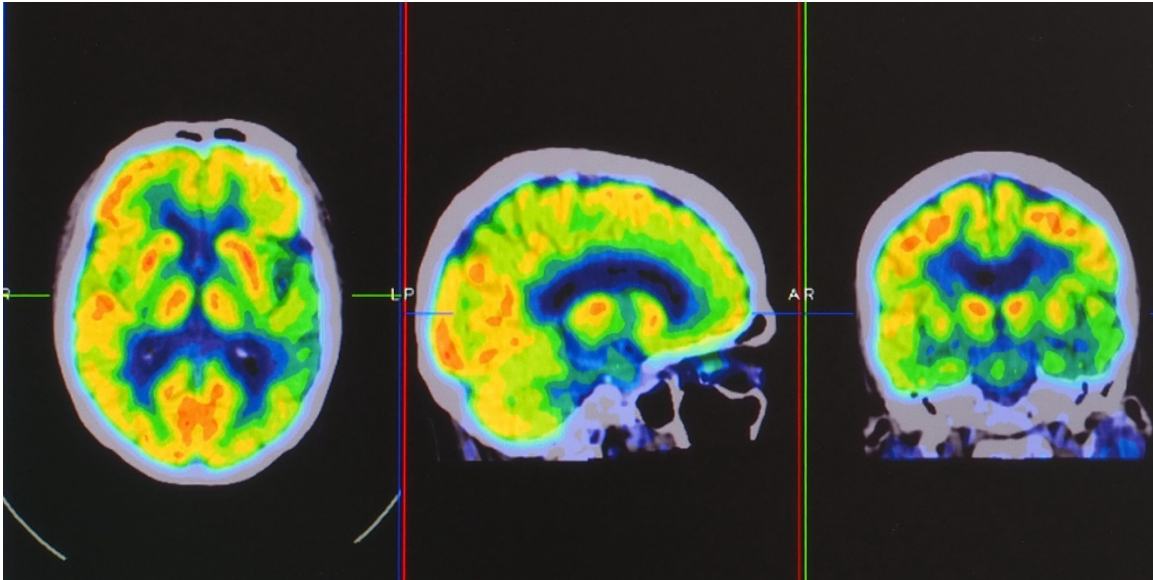
- **Computerized Axial Tomography (CAT):** an X-ray source rotates around the brain.
- Low-level X-ray beams pass through the brain and are picked up by a detector positioned on the opposite side.



- Two- and three dimensional views of the underlying tissues are then constructed
- Typically yields an image of a single cross section of the brain.

AI as a Tool for Neuroscience:- Brain Recording Techniques

- **Positron Emission Tomography (PET):** measure blood flow in the brain while a participant is carrying out a cognitive task.

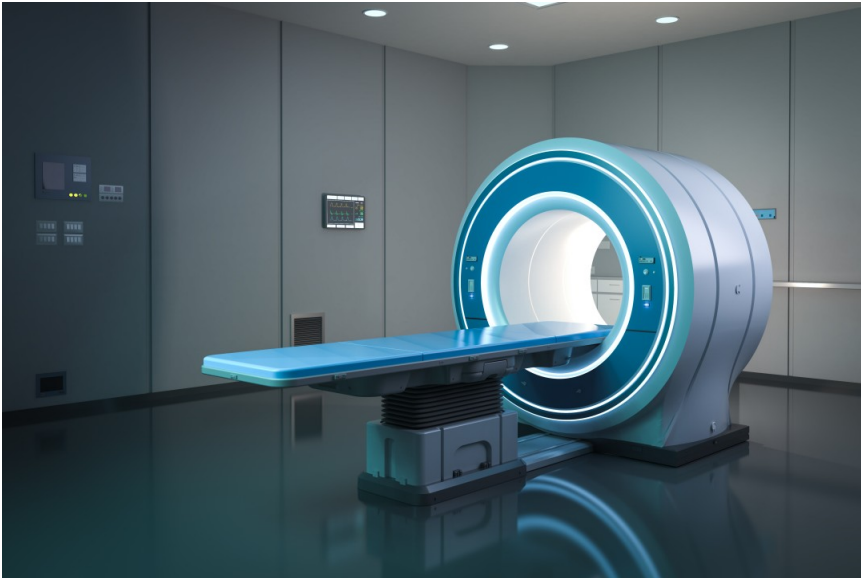


Cross-sectional view of a human brain across three planes: axial (left), sagittal (middle), and coronal (right).

- Brain areas that are more active during the carrying out of the task will show greater regional cerebral blood flow and, correspondingly, a greater concentration of the tracer molecules bearing the isotopes.
- Red/Orange/Yellow: Indicate areas of high metabolic activity or increased tracer uptake.
- Green/Blue: Indicate areas of lower metabolic activity.

AI as a Tool for Neuroscience:- Brain Recording Techniques

- Magnetic Resonance Imaging (MRI)



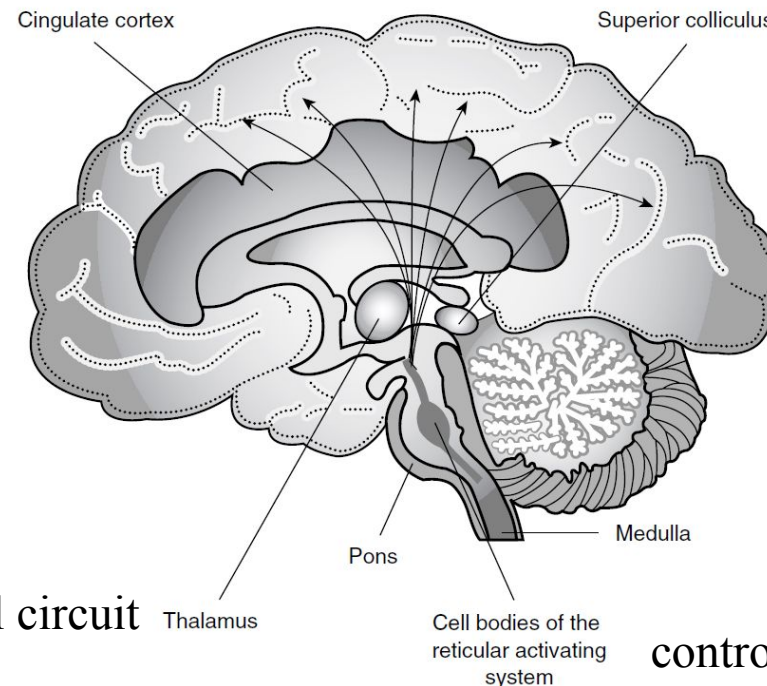
- A patient is placed inside a tube that contains a powerful magnet.
- A recently developed variant of this procedure is called **functional magnetic resonance imaging**, or fMRI for short, and is used to reveal changes in brain activity over time.

- In neuroimaging, AI has revolutionized image preprocessing and analysis.
- DL models, particularly CNNs, have automated tasks like segmenting brain structures, detecting subtle abnormalities, and identifying disease biomarkers.

The Neuroscience of Attention

- Attention plays a key role in many different cognitive phenomena.
- There are at least six distinct brain structures that underlie attentional effects
- We can think of these structures as being interconnected and as collectively forming an attentional neural network.

The site where a response is selected



Its function seems to be the shifting of one's visual attention from one object or spot in one's visual field to another

part of the general RAS arousal circuit

controls the brain's overall arousal and alertness levels

The Neuroscience of Attention

- Models of Attention:
 - A **Component Process Model**: each of the multiple brain areas responsible for the control of attention performs a distinct operation.
 - Specifically describes the changes that occur in visual selective attention, where **attention is shifted from one spatial location to another**.
 - The data that support this model come from an experimental paradigm
 - **Distributed Network Models**: an alternative model for the neural machinery that controls attention
 - The separate neural structures **are not specialized and functionally independent**, as they are in the component process model.

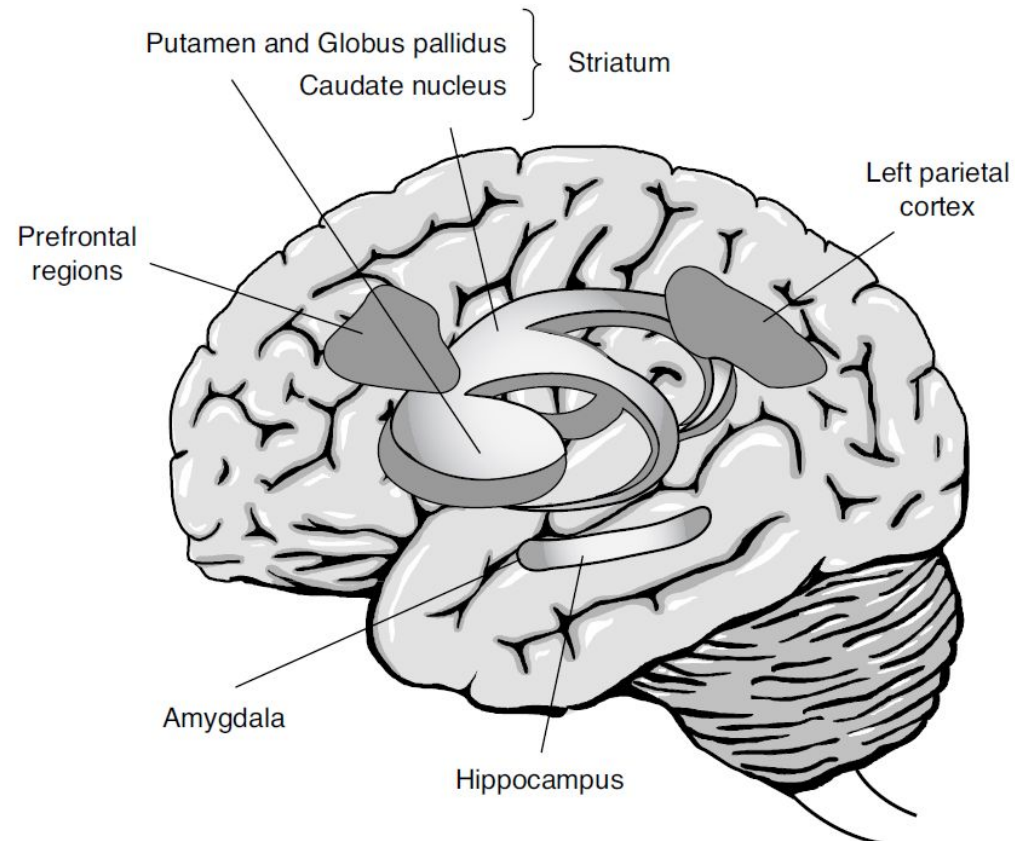
The Neuroscience of Memory

- Early neuroscience research on memory asked a very basic question: **Where in the brain are memories located?**
- Karl Lashley (1950) was searching for the engram, **a physical change in a part of the brain that is associated with learning.**
- Lashley's rationale was that if you destroy the part of the brain that contains the engram for the task, the monkey's memory of how to perform it will have been lost.
- In one experiment, he destroyed larger and larger areas of cortex in a graded manner.

The Neuroscience of Memory

- He found that the greater the amount of tissue that had been destroyed, the greater was the amount of time needed to retrain the monkeys to perform the task.
- Lashley concluded that memories are not laid down in any one area of the brain, but that all parts of the brain participate in memory storage... a principle referred to as **equipotentiality**.
- It suggests that multiple copies of a single memory are stored throughout the brain and that multiple brain regions participate in memory formation.

The Neuroscience of Memory



Brain areas that form part of the human memory system

Learning and Memory

- **Learning** has taken place when an event has provoked a nervous system alteration that in turn provokes a change in the organism's behavior.
- It is this change in the nervous system that is a memory.
- From a neuroscience point of view, a memory is essentially the formation of a neural circuit—the forging of a pathway among neurons.
- The memory is then contained within the circuit.

Learning and Memory

- Nervous systems preserve the past and help organisms to adapt to their environments.
- Learning requires **synaptic plasticity**, which refers to a capacity for change in the structure or biochemistry of a synapse.
- Donald Hebb (1949) **was perhaps the first to understand this and formulated what has become known as the Hebb rule.**
- The specific neural change that mediates this Hebbian form of learning is called long-term potentiation (LTP).
- LTP occurs in parts of the brain where learning and the creation of new neural circuits are important.

Computational Models in Neuroscience

- Computer modeling of neuronal, glial, synaptic and network function has become increasingly popular during the last few decades.
- Given the complexity of the nervous system, these models, especially the neuronal models have inherent deficiencies and are often criticized for their **oversimplification and limited applicability**
- A multi-level approach is suggested for bringing these models near to what exists in reality at cellular, receptor, genetic, molecular and network level.
- Nonlinear dynamical models have significantly contributed to the general understanding of brain activity as well as brain disorders.
- Data Driven and model based approach

The Neuroscience of Executive Function and Problem Solving

- **Executive function** refers to cognitive operations such as planning, the sequencing of behavior, the flexible use of information, and goal attainment.
- Many of these same operations are called upon in problem solving.
- Problem solving consists of trying to attain a final goal, which is the problem's solution, via the performance of a sequence of operations that leads to the attainment of individual subgoals.
- The hallmark of frontal lobe damage is a diminished capacity to perform goal-directed behaviors.

Problems that we should be able to solve in the next 50 years:

- How do circuits of neurons compute?
- What causes psychiatric and neurological illness?
- How do learning and memory work?
- Why do we sleep and dream?
- How do we make decisions?
- How does the brain represent abstract ideas?

Adolphs R. The unsolved problems of neuroscience. Trends Cogn Sci. 2015 Apr;19(4):173-5. doi: 10.1016/j.tics.2015.01.007. Epub 2015 Feb 20. PMID: 25703689; PMCID: PMC4574630.

Problems we may never solve

- How does the human brain compute?
- How can cognition be so flexible and generative?
- How and why does conscious experience arise?

Meta-question: How can a brain be built?

- Some argue that we can only understand the brain once we know how it could be built.
- Perhaps knowing how to build a brain will enable us to glean general principles that cut across the many individual questions, and across species.

Biocomputing

- Advances in neuroscience are paving the way for the construction of biological computers using neurons and brain tissue.
- The development of organoid intelligence (OI) has been made possible by three technological breakthroughs: electrophysiology, AI and cerebral organoids.
- OI promises to bring the two systems closer together, by reproducing the computational performance of the best computers and the energy efficiency of a human brain.



[Source](#): Polytechnique insight

Thank You!